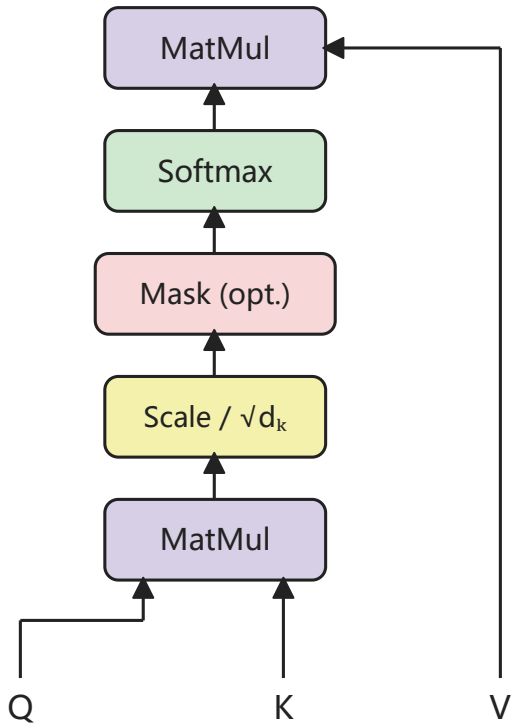
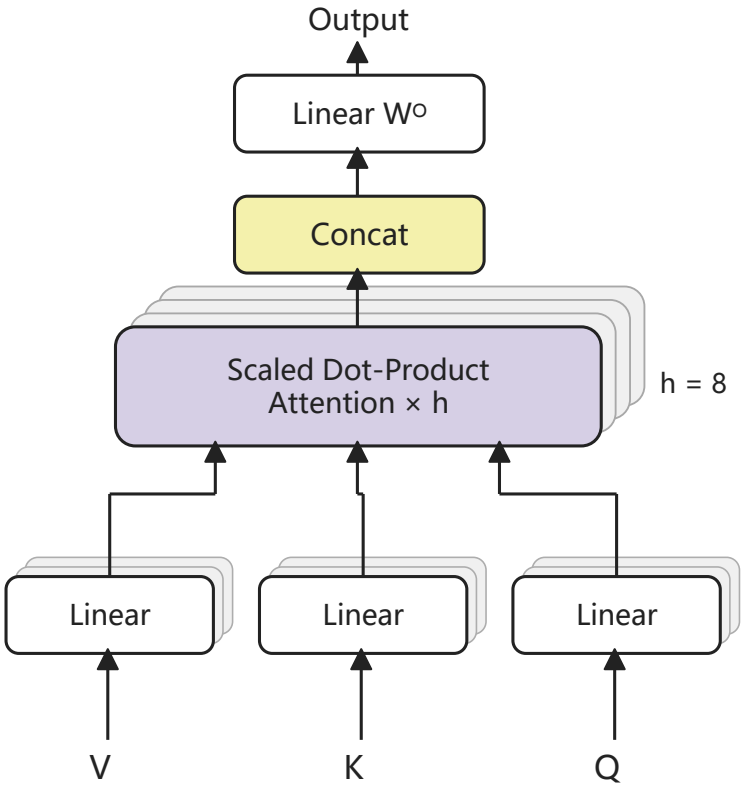
Scaled Dot-Product Attention



Multi-Head Attention



$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d_k} + M)V$$

每个头使用独立 Q/K/V 投影，结果拼接后再线性映射。

M 为可选掩码；被屏蔽位置在 Softmax 前加 −∞。

Vaswani et al. (2017), Fig. 2 · source-matched structural redraw